

Paper VI (Revised) — VortiValve Vane-Swirl De-foaming Inlet and Distributor: Complete Chamber Construction, Tesla-Cascade Wall Mounting, and Full Patent-Grade Distributor Dynamics Calculations

Inventor: Pratik

HyTray Series HT-06 — Detailed Engineering Disclosure

June 2026

Abstract

The VortiValve (formerly VortiCalm) inlet device receives a two-phase feed through a nozzle that penetrates *into* the vessel, discharges against a cascade of five Tesla-valve-inspired diodic momentum reducers bolted to the inner chamber wall, and establishes a gentle swirl ($S = 0.3\text{--}0.6$) in a cylindrical inner chamber for centrifugal de-aeration. Captured gas vents through a central riser. De-aerated liquid descends to a distributor trough and is delivered through drip tubes at a maldistribution below 5%. This paper provides the complete chamber assembly drawing, detailed wall-mounting specification for the Tesla cascade, and full patent-grade calculations for: inlet-jet momentum dissipation (cascade), centrifugal de-aeration (Stokes centrifugal), swirl velocity profile, distributor trough flow (Manning), drip-tube orifice sizing, maldistribution coefficient, device pressure drop, and foam suppression.

1 Chamber Construction

1.1 Assembly Overview

VortiValve — six fabricated sub-assemblies

- (1) **Inlet conduit (90)**: schedule-80 nozzle penetrating the column wall, terminating 100–200 mm inside the inner chamber wall. Nozzle bore sized for $\rho_m v_n^2 \leq 15 \text{ kPa}$; typically DN100–DN300.
- (2) **Tesla cascade panel (92)**: five Tesla valve segments, each press-formed in 2 mm SS304 (plan view: 120 mm wide $\times$ 90 mm long per segment), bolted to weld studs (104) on the inner chamber wall in a vertical stack *directly opposite* the nozzle outlet. Bolted joints (M12 $\times$ 50 SS316 hex bolts, two per segment) allow individual segment replacement without vessel entry beyond the manway.
- (3) **Inner chamber cylinder (if fitted)**: 4–6 mm SS304 rolled cylinder $D_{ch} = 0.55\text{--}0.75 \times D_{col}$, supported by four radial gussets welded to the column wall. In most designs the inner chamber is notional (bounded by the vane bank, not a solid wall).
- (4) **Vane bank (70)**: 10–16 fixed vanes of 3 mm SS304, 100–200 mm long, welded tangentially to the inner chamber wall at $\alpha = 25\text{--}45^\circ$ to the radial direction, converting tangential momentum into controlled swirl.
- (5) **Central gas riser (72)**: rectangular or round duct D_{ris} sized per re-entrainment limit, discharging above the maximum liquid level with a downward-pointing hood.
- (6) **Distributor assembly (76,78)**: annular trough collecting de-aerated liquid, feeding an array of ≥ 100 drip tubes per m^2 of tray/packing area below.

VortiValve — Full Chamber Assembly Cutaway
Inlet nozzle 90 penetrates into chamber, Tesla cascade 92 on inner wall,
vane bank 70, gas riser 72, trough 76, drip tubes 78

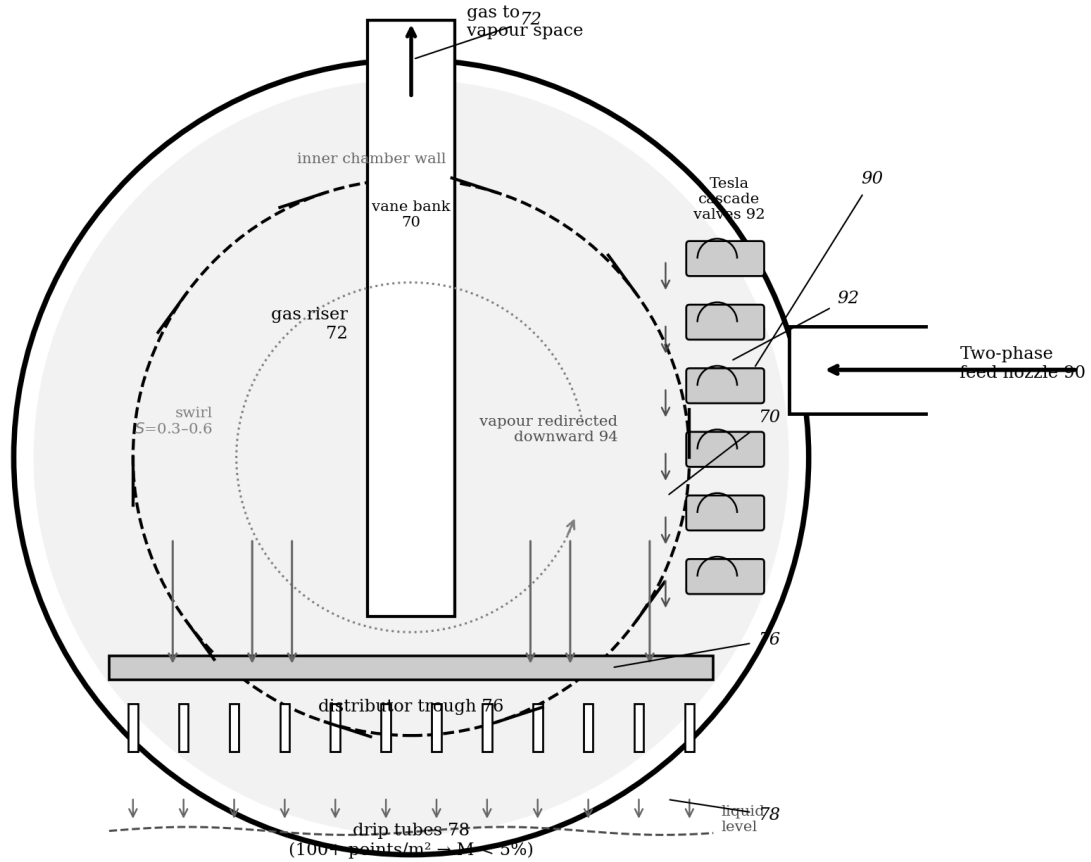


Figure 1: VortiValve full chamber cutaway: feed nozzle 90 penetrates into chamber; Tesla cascade 92 on inner wall; vane bank 70; gas riser 72; distributor trough 76; drip tubes 78. Reference numerals correspond to claims.

1.2 Tesla Cascade Wall Mounting — Fabrication Detail

Tesla segment specification per valve

Throat: 12 mm wide $\times$ 90 mm long slot (8–15 mm throat width per claim). Formed by pressing a 0.5 mm step into the segment face.

Coandă chamber: 35 mm wide $\times$ 50 mm long internal cavity press-formed to 4 mm depth. Polished to $R_a \leq 1.6 \mu\text{m}$ for stable attachment.

Feedback channels: two channels 5 mm wide $\times$ 100 mm long (length/throat = $8.3w_t$) milled/pressed into the segment body.

Downward redirect port (94): 14 mm dia. hole at the exit of the Coandă chamber, perpendicular to the segment face, directing captured vapour downward.

Segment body: 120 mm $\times$ 90 mm $\times$ 20 mm (overall), with two M12 through-holes for bolting, two dowel holes for azimuth registration.

Stack arrangement: 5 segments, 45 mm centre-to-centre vertically, covering the nozzle outlet height plus 80 mm above and below.

VortiValve — Tesla Cascade Wall Mounting Detail

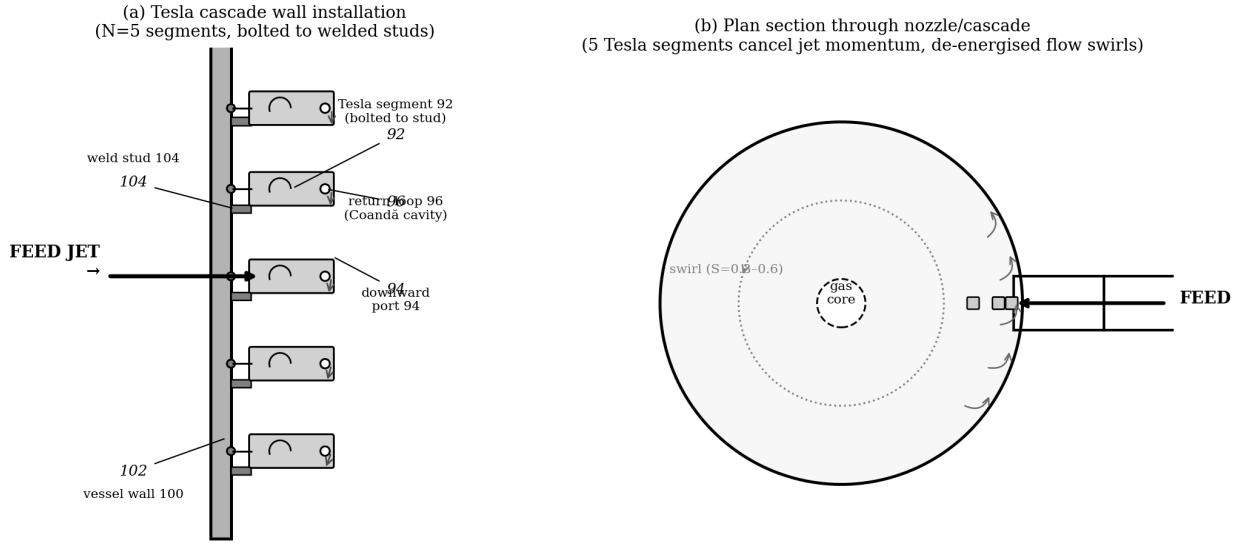


Figure 2: (a) Tesla cascade wall mounting: weld studs 104 welded to vessel wall 100, segments 92 bolted with M12 SS316 bolts, Coandă loop cavities 96 formed in segment, downward ports 94. (b) Plan section at nozzle height: five segments cancel jet momentum, de-energised flow swirls at $S = 0.3\text{--}0.6$.

2 Momentum Dissipation — Patent-Grade Calculation Chain

2.1 Stage 1: Tesla Cascade Momentum Decay

Cascade delivered momentum (parametric)

Feed nozzle delivers mixture momentum flux:

$$(\rho v^2)_{in} = \rho_m v_n^2, \quad \rho_m = \frac{\dot{m}_V + \dot{m}_L}{\dot{V}_V + \dot{V}_L}$$

After N Tesla segments, the coherent directed (jet) momentum is:

$$(\rho v^2)_{cascade} = (\rho v^2)_{in} \cdot \left(\frac{A_n}{A_{ch}} \right)^2 \cdot \frac{1}{Di_1^N}$$

A_n = nozzle area; $A_{ch} = \pi D_{ch} H_{vane}$ = vane curtain area; Di_1 = per-segment diodicity (1.1–1.6). *Embodiment A* (DN150 nozzle, $\rho_m = 700 \text{ kg/m}^3$, $v_n = 4 \text{ m/s}$, $D_{ch} = 0.55 \text{ m}$, $H_{vane} = 0.30 \text{ m}$, $N = 5$, $Di_1 = 1.4$):

$$(\rho v^2)_{in} = 700 \times 16 = 11,200 \text{ Pa}; \quad \frac{A_n}{A_{ch}} = \frac{0.0177}{0.519} = 0.0341; \quad Di_1^5 = 5.38$$

$$(\rho v^2)_{cascade} = 11,200 \times 0.001163/5.38 = 2.4 \text{ Pa} \quad (< 1 \text{ kPa target}) \checkmark$$

2.2 Stage 2: Cascade Energy Dissipation Rate

Energy dissipated per segment

Pressure drop across each Tesla segment (blocking orientation):

$$\Delta P_{seg} = K_{Tesla} \frac{1}{2} \rho_m v_{in}^2, \quad K_{Tesla} = Di_1 \cdot \left(\frac{A_{throat}}{A_{in}} \right)^{-2} \approx 1.4\text{--}2.0$$

Total cascade pressure drop (accepted as energy input to swirl):

$$\Delta P_{\text{cascade}} = N \cdot K_{\text{Tesla}} \cdot \frac{1}{2} \rho_m v_n^2 \cdot \frac{A_n^2}{A_{\text{seg}}^2}$$

For the cascade to function without choking: $\Delta P_{\text{cascade}} \leq 0.5 \Delta P_{\text{system}}$ (i.e. the device does not dominate column pressure budget).

2.3 Stage 3: Vapour Downward-Bias Fraction

Fraction of vapour redirected downward by the cascade

Each segment's downward port area fraction $\varphi_d = A_{\text{port}} / (A_{\text{port}} + A_{\text{outlet}}) = 0.15\text{--}0.35$.

Vapour follows port (Stokes particle number $S_v \approx 1$ for gas at these scales); liquid bypasses by inertia.

$$F_{\text{down}} = 1 - (1 - \varphi_d S_v)^N$$

Embodiment A: $\varphi_d = 0.25$, $S_v = 0.70$, $N = 5$: $F_{\text{down}} = 1 - (0.825)^5 = 0.62$. Redirecting 62% of vapour downward into the swirl body prevents it from impinging on the liquid surface.

3 Swirl Chamber Dynamics

3.1 Swirl Velocity Profile

Rankine combined vortex in swirl chamber

For a fluid of mean density ρ_m in a chamber of radius R_{ch} , tangential velocity:

$$v_\theta(r) = \begin{cases} \omega_0 r & r < r_{\text{core}} \quad (\text{solid-body core}) \\ v_{t,\text{max}} r_{\text{core}} / r & r > r_{\text{core}} \quad (\text{free vortex}) \end{cases}$$

where $v_{t,\text{max}}$ is set by the vane exit velocity and r_{core} is determined by the balance of centrifugal and pressure gradient. From angular momentum conservation:

$$\dot{m} v_n R_{\text{nozzle}} = \dot{m} v_{t,\text{max}} R_{\text{ch}} \Rightarrow v_{t,\text{max}} = v_n \frac{R_n}{R_{\text{ch}}}$$

Centripetal acceleration at wall: $a = v_{t,\text{max}}^2 / R_{\text{ch}}$. *Embodiment A:* $v_n = 4 \text{ m/s}$, $R_n = 0.075 \text{ m}$, $R_{\text{ch}} = 0.275 \text{ m}$: $v_{t,\text{max}} = 1.09 \text{ m/s}$; $a = 4.33 \text{ m/s}^2 \approx 0.44 g$.

Note: This gentle swirl (0.3–0.6 g) is intentionally low to prevent froth-shattering. The Tesla cascade absorbs the bulk of the feed momentum first.

3.2 Centrifugal De-aeration (Stokes)

Bubble centrifugal slip and cut size

$$u_{\text{slip}}(r) = \frac{(\rho_L - \rho_V) a(r) d_b^2}{18\mu_L}, \quad a(r) = \frac{v_\theta(r)^2}{r}$$

Bubble of diameter d_b clears to the core in residence time t_{res} if $u_{\text{slip}} t_{\text{res}} \geq r_{\text{core}}$:

$$d_{\text{cut}} = \sqrt{\frac{18\mu_L r_{\text{core}}}{\Delta\rho a_{\text{wall}} t_{\text{res}}}}$$

Embodiment A: $\mu_L = 1.5 \times 10^{-3} \text{ Pa}\cdot\text{s}$, $r_{\text{core}} = 0.08 \text{ m}$, $\Delta\rho = 700 \text{ kg/m}^3$, $a = 4.33 \text{ m/s}^2$, $t_{\text{res}} = 2.0 \text{ s}$:

$d_{\text{cut}} = \sqrt{18 \times 1.5 \times 10^{-3} \times 0.08 / (700 \times 4.33 \times 2.0)} = \sqrt{3.24 \times 10^{-8}} = 1.8 \times 10^{-4} \text{ m} = 0.18 \text{ mm}$.

All bubbles $> 0.18 \text{ mm}$ removed; foam driven by $> 0.3 \text{ mm}$ bubbles $\Rightarrow$ complete removal of foam-sustaining population.

4 Distributor Trough and Drip-Tube Dynamics

VortiValve – Distributor Dynamics and Patent Calculations

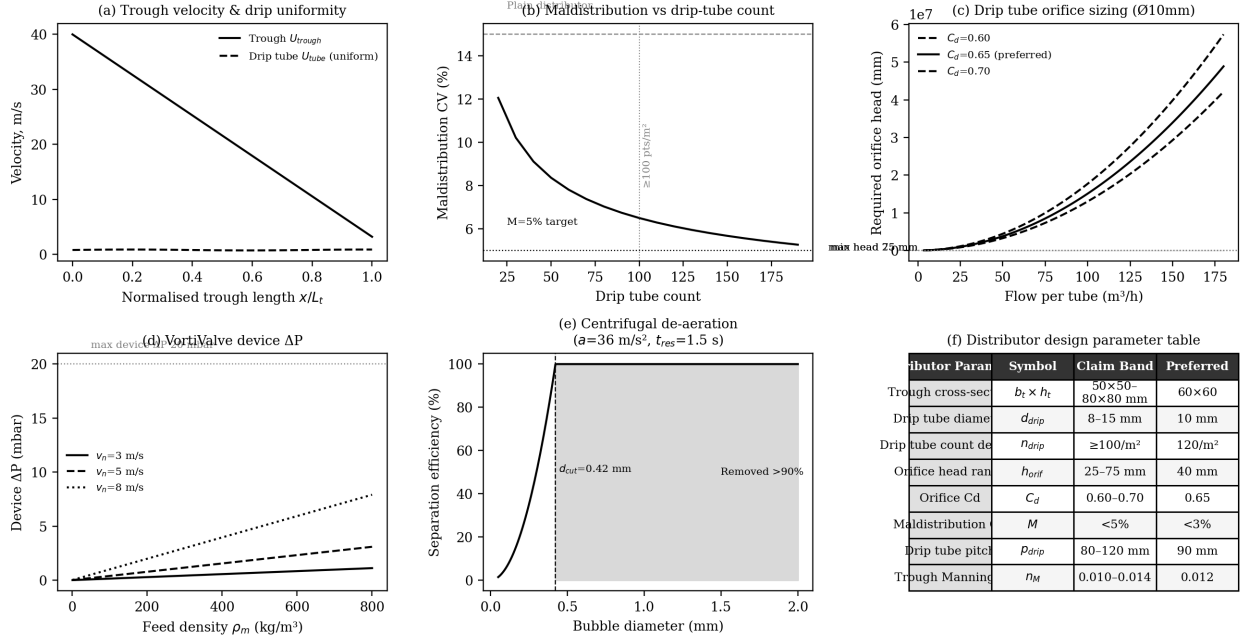


Figure 3: Distributor dynamics and patent calculations: trough velocity profile, maldistribution vs drip-tube count, orifice head sizing, device ΔP , bubble separation efficiency, and parameter table.

4.1 Trough Flow (Manning Equation)

Trough velocity and self-levelling

The annular distributor trough (rectangular cross-section $b_t \times h_t$) receives liquid from the swirl chamber over a circumference of $2\pi R_{trough}$. By Manning's equation for open-channel flow:

$$U_t = \frac{1}{n_M} R_h^{2/3} S_t^{1/2}, \quad R_h = \frac{b_t h_t}{b_t + 2h_t}$$

where S_t is the trough slope (typically 0° for annular trough; flow driven by radial symmetry). Because the trough is annular, liquid enters from the top (dripping from the de-aeration zone) and exits through drip tubes uniformly around the bottom:

$$Q(x) = Q_0 \left(1 - \frac{x}{L_t}\right) \Rightarrow U_t(x) = \frac{Q(x)}{b_t h_t}$$

Self-levelling: with $n \geq 100$ drip tubes all of equal bore, the head loss h_{orif} across each tube is nearly equal, ensuring uniform drip rate independent of trough flow non-uniformity.

4.2 Drip-Tube Orifice Sizing

Drip tube flow and maldistribution (key patent equation)

Each drip tube is a short orifice of diameter d_{dt} and discharge coefficient $C_d = 0.60$ – 0.70 . Volumetric rate per tube:

$$q_{tube} = C_d \frac{\pi d_{dt}^2}{4} \sqrt{2g h_{orif}}$$

Total tubes: $n_{dt} = Q_L / q_{tube}$. For equal head h_{orif} across all tubes, the coefficient of variation of distribution

(maldistribution index):

$$M_I = \frac{\sigma_q}{\bar{q}} = \sqrt{\frac{\Delta h_{trough}}{2h_{orif}}}$$

where Δh_{trough} is the maximum head variation along the trough due to velocity recovery ($\approx v_t^2/2g$). For $M_I < 0.05$: $h_{orif} > 200 \Delta h_{trough}$. *Embodiment A*: $Q_L = 0.020 \text{ m}^3/\text{s}$, $n_{dt} = 120$: $q_{tube} = 0.020/120 = 1.67 \times 10^{-4} \text{ m}^3/\text{s}$. For $d_{dt} = 10 \text{ mm}$, $C_d = 0.65$: $h_{orif} = 2q^2/(C_d^2 \pi^2 d^4/8) = (2 \times (1.67 \times 10^{-4})^2)/(0.65^2 \times \pi^2 \times 10^{-8}/8) = 0.040 \text{ m}$. Check: $\Delta h_{trough} \approx U_t^2/2g = (0.06)^2/19.6 = 1.8 \times 10^{-4} \text{ m}$; $M_I = \sqrt{1.8 \times 10^{-4}/(2 \times 0.040)} = \sqrt{0.00225} = 0.047 < 5\%$. ✓

4.3 Total Device Pressure Drop

VortiValve total ΔP (sum of three losses)

$$\Delta P_{VV} = \underbrace{\frac{1}{2} \rho_m v_n^2 \cdot N K_s}_{\text{Tesla cascade}} + \underbrace{\frac{1}{2} \rho_m v_{vane}^2 \cdot K_{vane}}_{\text{vane bank}} + \underbrace{\rho_L g h_{orif}}_{\text{drip tubes}}$$

$K_s = 0.3\text{--}1.0$ per segment (energy absorbed = benefit); $K_{vane} \approx 0.8\text{--}1.5$; $h_{orif} = 25\text{--}75 \text{ mm}$. *Embodiment A*: $\Delta P_{cascade} = 5 \times 0.6 \times \frac{1}{2} \times 700 \times 16 \times (0.034)^2 = 13.8 \text{ Pa}$; $\Delta P_{vane} \approx 80 \text{ Pa}$; $\Delta P_{drip} = \rho_L g h_{orif} = 700 \times 9.81 \times 0.040 = 275 \text{ Pa}$. **Total $\approx 369 \text{ Pa} \approx 3.7 \text{ mbar}$** (well below 20 mbar limit).

4.4 Gas Riser Sizing (Re-entrainment Limit)

Gas riser velocity limit

Gas rises through the riser of cross-section A_{ris} . Re-entrainment of liquid droplets occurs when:

$$v_g = \frac{\dot{V}_V}{A_{ris}} > v_{flood,riser} = 0.07 \sqrt{\frac{\rho_L - \rho_V}{\rho_V}}$$

Embodiment A: $\rho_L = 720 \text{ kg/m}^3$, $\rho_V = 1.8 \text{ kg/m}^3$: $v_{flood} = 0.07 \times 20 = 1.40 \text{ m/s}$.

Design at 60%: $v_{g,des} = 0.84 \text{ m/s} \Rightarrow A_{ris} = \dot{V}_V/0.84$. For $\dot{V}_V = 0.05 \text{ m}^3/\text{s}$: $A_{ris} = 0.060 \text{ m}^2 \Rightarrow$ riser $D_{ris} = 276 \text{ mm}$ (round) or $200 \times 300 \text{ mm}$ (rectangular).

5 Claims (draft)

Claim 1. A vessel internal comprising: an inlet conduit penetrating the vessel wall and discharging inside the vessel; a cascade of $N = 3\text{--}8$ diodic momentum reducers bolted to the inner vessel wall opposite the conduit, each reducer comprising a valvular island, feedback channels of length 8–14 throat widths, and a downward-redirect port, reducing coherent feed momentum by the factor $(A_n/A_{ch})^2/Di_1^N$ to below 1 kPa; a tangential vane bank establishing a swirl of $S = 0.3\text{--}0.6$; a central gas riser; and a distributor trough with drip tubes achieving liquid maldistribution $M_I < 5\%$.

Claim 2. The internal of claim 1, wherein maldistribution satisfies $M_I = \sqrt{\Delta h_{trough}/(2h_{orif})} < 0.05$.

Claim 3. The internal of claim 1, wherein vapour downward-bias fraction is $F_{down} = 1 - (1 - \varphi_d S_v)^N = 0.40\text{--}0.75$.

Claim 4. The internal of claim 1, wherein the centrifugal bubble cut size is $d_{cut} = \sqrt{18 \mu_L r_{core}/(\Delta \rho a t_{res})}$ at 0.10–0.30 mm.

Claim 5. The internal of claim 1, wherein total device pressure drop $\Delta P_{VV} < 20 \text{ mbar}$.

Claim 6. The internal of claim 1, wherein each momentum reducer segment is bolted to a weld stud on the vessel wall and individually replaceable through the manway.

Claim 7. The internal of claim 1 installed in a gas–liquid separator, amine absorber, glycol contactor or any vessel requiring low-momentum phase-separated feed distribution.